

Practice Problem Set 1

ECON 311 — Intermediate Macroeconomics — Fall 2026

Problem 1 *Growth-rate arithmetic.*

Suppose x , y , and w grow at constant rates g_x , g_y , and g_w , respectively. Express the growth rate of z in terms of these growth rates for each of the following:

(a) $z = x^2 y^3$

(b) $z = \frac{x^{0.6}}{y^{0.4}}$

(c) $z = x^{-0.5} \left(\frac{1}{w} \right)^{0.3} y^{0.8}$

(d) $z = \left(\frac{xy}{w^2} \right)^{1/2}$

Problem 2 *Returns to scale.*

For each production function below, determine whether it exhibits *increasing*, *constant*, or *decreasing* returns to scale in capital K and labour L . (Treat $\bar{A}$ as a fixed positive constant throughout.) Show your work clearly.

(a) $Y = \bar{A} K^{1/4} L^{3/4}$

(b) $Y = K^{0.6} L^{0.6}$

(c) $Y = K^{1/3} L^{2/3} + \bar{A}$

(d) $Y = K + K^{0.5} L^{0.5}$

Problem 3 *National income accounting — three methods.*

The economy of **Silverdale** can be described by the following activities during one year:

- A **lumber company** hires workers for \$18,000 in wages, harvests 500 cubic metres of timber, and sells 400 cubic metres to a furniture manufacturer at \$60 per cubic metre. The remaining 100 cubic metres are milled into finished planks and sold directly to households for \$9,000.
 - A **furniture manufacturer** purchases 400 cubic metres of timber at \$60 per cubic metre, pays its workers \$30,000 in wages, and produces 50 chairs. Of those chairs: 35 are sold to households at \$1,200 each, 10 are sold to a government-run community centre at \$1,200 each, and 5 are exported at \$1,400 each.
- (a) Calculate GDP using the **expenditure approach**. Identify each component (C , I , G , NX) clearly.
- (b) Calculate GDP using the **income approach**. Identify all labour income and capital (profit) income.
- (c) Calculate GDP using the **value-added approach**. Show the value added at each stage of production.
- (d) Verify that all three methods give the same answer. If they do not, identify and correct any error in your work.

Problem 4 *Real GDP, nominal GDP, and the GDP deflator.*

Use the Silverdale economy from Problem 3 as Year 1. In Year 2, the same quantities of all goods are produced and sold, but:

- The price of timber rises to \$75 per cubic metre.
 - The price of a finished plank bundle rises to \$11,500 (for the 100-cubic-metre lot).
 - The price of a chair sold domestically or to government rises to \$1,500; the export price rises to \$1,700 per chair.
- (a) What is nominal GDP in Year 2? (Hint: use the expenditure approach on final goods only.)
- (b) What is *real* GDP in Year 2, using Year 1 prices as the base year? What is the real

GDP growth rate between Year 1 and Year 2?

- (c) Calculate the GDP deflator for Year 2 (with Year 1 as base year). What is the percentage change in the general price level?
- (d) Explain briefly why real GDP growth and nominal GDP growth differ in this example.

Problem 5 *Economic convergence.*

In the year 2000, **Country Alder** has GDP per capita of \$5,000 and grows at a constant rate of 4% per year. **Country Birch** has GDP per capita of \$24,000 and grows at 2% per year.

- (a) Derive a formula for the year t^* at which Country Alder catches up to Country Birch. Solve for t^* numerically.
- (b) Suppose that, in the first 15 years, Country Alder instead grows at 7% (due to institutional reforms), and then reverts to its original 4% growth rate thereafter. How many years after the year 2000 does it take Country Alder to catch up with Country Birch under this scenario?
- (c) Give a brief economic interpretation of why the catch-up time changes. What real-world mechanism might allow Country Alder to grow at 7% in the early period?

Problem 6 *From aggregate to per-capita variables.*

Consider a closed economy with no government sector. The aggregate production function is:

$$Y_t = A K_t^\alpha L_t^{1-\alpha}, \quad 0 < \alpha < 1.$$

The population grows at rate $\bar{n}$: $L_{t+1} = (1 + \bar{n})L_t$. Households save a fixed fraction $\bar{s}$ of output each period: $I_t = \bar{s} Y_t$. Capital depreciates at rate δ :

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

Lowercase letters denote per-person quantities: $k_t = K_t/L_t$, $y_t = Y_t/L_t$.

- (a) Show that $y_t = A k_t^\alpha$.
- (b) Starting from the capital accumulation equation above, derive the per-capita capital

accumulation equation:

$$\Delta k_{t+1} = \bar{s} y_t - (\delta + \bar{n}) k_t.$$

Show every algebraic step clearly.

- (c) Interpret each term on the right-hand side of the equation derived in part (b) economically. What does each term represent?

Problem 7 *Solow model: steady state.*

Use the general setup from Problem 6 with $\alpha = 1/3$.

- (a) Find the steady-state capital per capita k^* and output per capita y^* in terms of the parameters A , $\bar{s}$, δ , and $\bar{n}$.
- (b) Suppose the savings rate $\bar{s}$ increases. What happens to k^* and y^* ? Use the formulas from part (a) to explain, and give an economic interpretation.
- (c) Suppose the population growth rate $\bar{n}$ increases. What happens to k^* and y^* ? Provide an economic explanation for the direction of the effect.

Problem 8 *Solow model: numerical dynamics.*

Use the per-capita capital accumulation equation from Problem 6(b) with $\alpha = 1/3$ and the following parameter values:

$$A = 1, \quad \bar{s} = 0.25, \quad \delta = 0.06, \quad \bar{n} = 0.02, \quad k_1 = 6.$$

- (a) Calculate k^* and y^* using your formulas from Problem 7(a).
- (b) Calculate Δk_2 , Δk_3 , Δk_4 , and Δk_5 . Show your work for at least the first two periods.
- (c) Describe the pattern you observe in the sequence $\Delta k_2, \Delta k_3, \Delta k_4, \Delta k_5$. Explain *why* this pattern arises.
- (d) Is the economy above or below its steady state at period 1? How can you tell from your calculations?

Problem 9 *Solow diagram.*

Using the parameter values from Problem 8, draw a fully labelled **Solow diagram** with k on the horizontal axis.

- (a) Label the two curves, their intercepts (if any), and the steady-state capital level k^* .
- (b) Mark the initial position $k_1 = 6$ on the diagram.
- (c) Draw an arrow or arrows indicating the direction of movement of k_t from k_1 toward the steady state.
- (d) What would happen to the diagram if the depreciation rate δ increased? Describe which curve(s) shift and in what direction, and state the new qualitative effect on k^* .

Problem 10 *Comparative statics and policy analysis.*

Return to the general Solow model from Problems 6 and 7.

- (a) A government implements a policy that raises the savings rate permanently from $\bar{s}$ to $\bar{s}' > \bar{s}$. Starting from the old steady state k^* , describe the *transitional dynamics*: what happens to Δk_{t+1} immediately after the policy, and what happens as the economy converges to the new steady state k'^* ?
- (b) Using the expression for y^* from Problem 7(a), show algebraically that a country with a *higher* savings rate will have a *higher* level of output per capita in the steady state, everything else equal.
- (c) The Solow model predicts *conditional convergence*. Explain what this means. How does it differ from *absolute convergence*? Use the steady-state formula for y^* in your answer.
- (d) In the Solow model without technological progress, the long-run growth rate of output per capita is zero. Explain why. Does this seem consistent with the real-world experience of developed economies? Briefly discuss one extension to the model that can generate sustained per-capita growth.

End of Practice Problem Set 1

Instructor solutions are available separately.